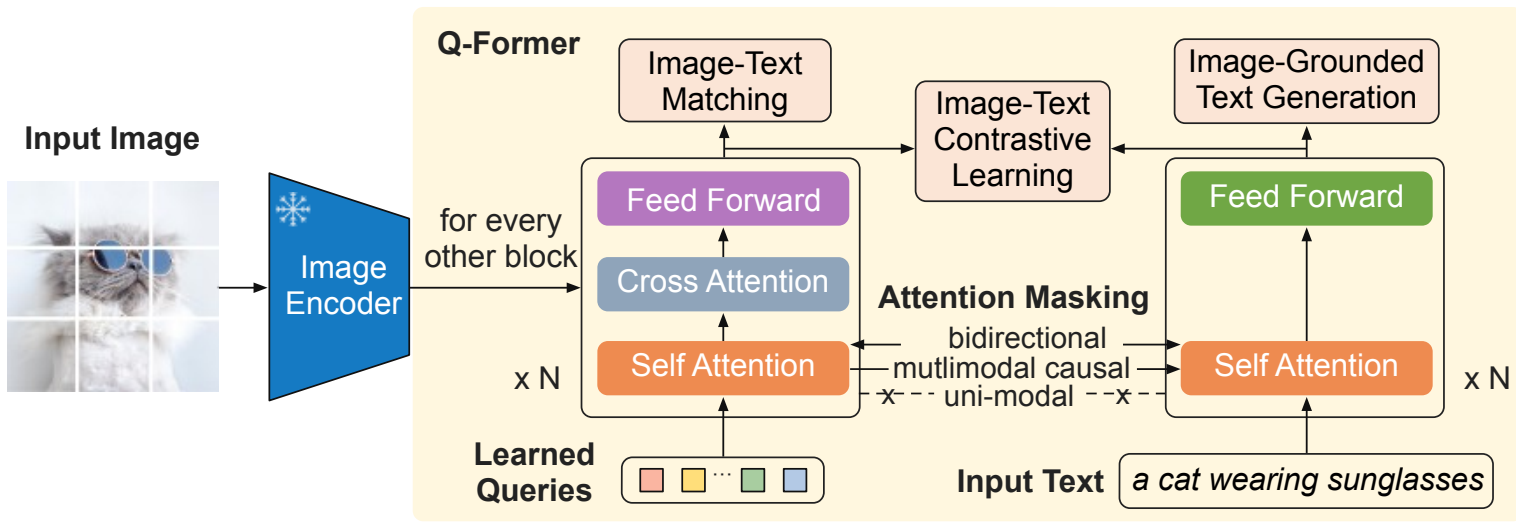
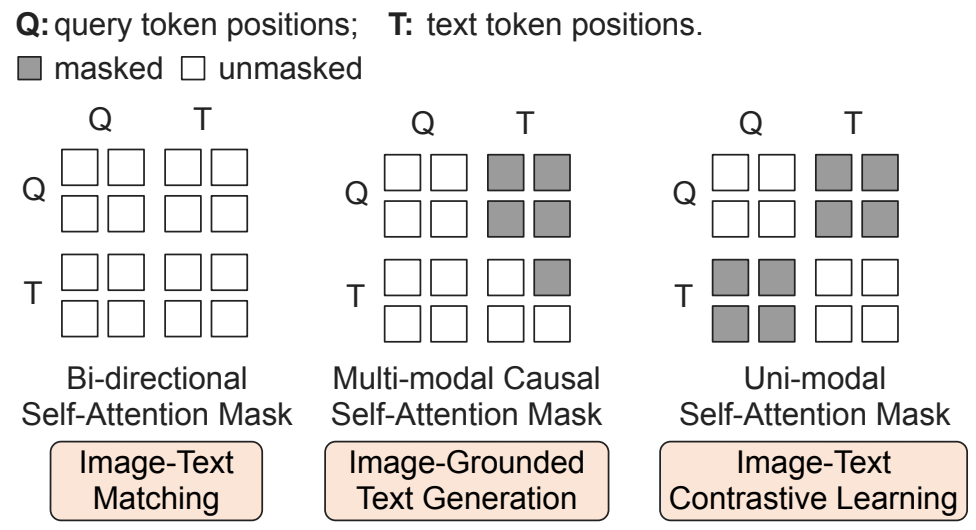

Input Image
Image Encoder
Q-Former
for every other block
x N
Image-Text Matching
Image-Text Contrastive Learning
Image-Grounded Text Generation
Feed Forward
Cross Attention
Self Attention
Attention Masking
bidirectional
mutlimodal causal
uni-modal
Feed Forward
Self Attention
x N
Learned Queries
Input Text
a cat wearing sunglasses


Q: query token positions; T: text token positions.
masked unmasked
Q
T
Bi-directional Self-Attention Mask
Image-Text Matching
Multi-modal Causal Self-Attention Mask
Image-Grounded Text Generation
Uni-modal Self-Attention Mask
Image-Text Contrastive Learning